Homework 4: Build and Honestly Test a Strategy

Data snapshot: 18 October 2026 | Index: ^SP500TR (total return)

1. Data and split

The requested 10 September 2011 was a Saturday, so the split uses the preceding trading close, 9 September. Training has **5,469 closes** with mean price **1,358.78**. The test input has **3,857 closes** including the 60 training closes used only for warmup; its mean is **6,875.45**. Its first earned return is 9 to 12 September 2011.

2-5. Backtester and controls (training)

The 60-close window ends at today's close; the decision applies only to the return to the next close. The index and strategy both start at 1 over the same traded dates. The random rule uses seed 0.

Rule	Total return	Annualised return
Always hold / index	+432.27%	+8.10%
Always cash	+0.00%	+0.00%
Random hold, 70%	+361.36%	+7.38%
Class: hold unless 5-day return positive	+206.70%	+5.36%



The always-hold and index curves overlap exactly; always cash returns exactly zero.

6-7. All variants tried and final rule

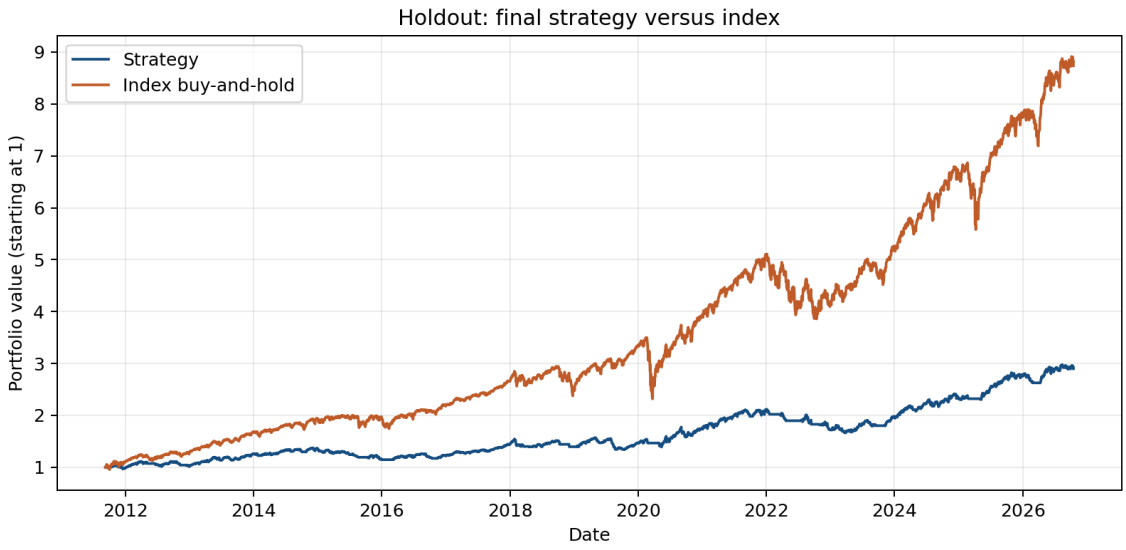
Training rule	Annualised
Close above 60-day moving average	+5.72%
Close above 30-day moving average	+3.84%
Positive 30-day return	+3.28%
Nonpositive 15-day return	+3.43%
Above 60-day average AND nonpositive 5-day return	+2.52%

I selected the first rule on training data: hold when the latest close is above its 60-day average; otherwise hold cash. A trend filter could work beyond this sample if sustained declines tend to persist, though whipsaws can make the rule miss recoveries and generate trading costs.

8. One holdout evaluation

The rule was locked before evaluating the holdout and was not changed afterward. The 60 warmup closes generate no test returns; results cover 3,797 subsequent trading returns.

Holdout metric	60-day strategy	Buy-and-hold index
Total return	+189.81%	+773.51%
Annualised return	+7.32%	+15.47%
Maximum drawdown	-21.40%	-33.79%
Fraction invested	76.32%	100.00%



9. Latest signal

Using the latest 60 closes through **16 October 2026**, the rule says **CASH**.

10. What the test means

The strategy earned 7.32% annualised in the holdout versus 5.72% in training, but lost plainly to buy-and-hold's 15.47% in the holdout; its lower drawdown is a real historical tradeoff, not evidence of superior overall performance. I am not surprised that a trend rule missed part of a strong bull market, although the size of the return gap is striking. I would not invest my own money based on this result: five rules were tried, and the public historical holdout is only one market path, not a genuinely new prospective sample. To trust it, I would want a pre-registered rule tested prospectively and across independent markets and regimes, with realistic costs, taxes, cash yield, and executable timing. The backtest assumes a trade at the same close used to compute the signal, which is optimistic, and it ignores slippage and trading costs; a total-return index also cannot itself be traded. The lesson is that a smoother ride can come with substantial opportunity cost, and a favorable training comparison among several variants is not proof of a durable edge.